

Analytical method interferences for perfluoropentanoic acid (PFPeA) and perfluorobutanoic acid (PFBA) in biological and environmental samples

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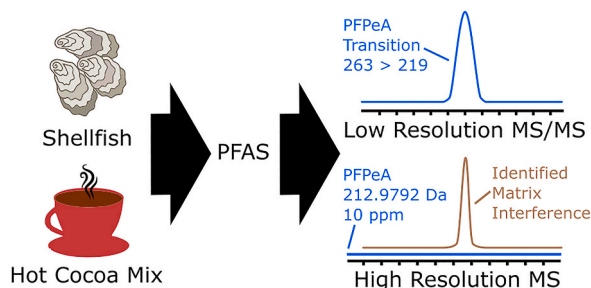
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HIGHLIGHTS

- PFPeA & PFBA have only one MS/MS transition on low resolution instruments.
- We identified a PFPeA interference using HRMS in shellfish at 263.1288 Da.
- We verified a separate PFPeA interferent at 262.1087 Da in hot cocoa.
- Authors suggest altering LC conditions & using matched IS when investigating short chain PFAS.

GRAPHICAL ABSTRACT



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ABSTRACT

While high-resolution MS (HRMS) can be used for identification and quantification of novel per- and poly-fluorinated alkyl substances (PFAS), low-resolution MS/MS is the more commonly used and affordable approach for routine PFAS monitoring. Of note, perfluoropentanoic acid (PFPeA) and perfluorobutanoic acid (PFBA), two of the smaller carboxylic acid containing-PFAS, have only one major MS/MS transition, preventing the use of qualitative transitions for verification on low-resolution instrumentation. Recently our lab has observed widespread chemical interference in the quantitative ion channel for PFPeA (263 → 219) and PFBA (213 → 169) in numerous matrices. PFPeA interference was investigated using HRMS and putatively assigned as a diprotic unsaturated fatty acid (263.1288 Da) in shellfish and a separate interferent (¹³C isotope of 262.1087 Da) in hot cocoa, which had been previously described by the FDA. PFBA interference caused by saturated oxo-fatty acids, previously demonstrated in tissue, was also observed in liquid condensate from a residential air conditioning unit. Therefore, in support of PFAS analysis on low-resolution instrumentation, authors recommend several adjustments to analytical methods including altering liquid chromatography (LC) conditions as well as using matched internal standards to investigate and expressly confirm PFBA and PFPeA detections in both biological and environmental samples.

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1. Introduction

Per- and polyfluorinated substances (PFAS) are a class of thousands of unique chemicals containing carbon-fluorine bonds.¹ Production of the first PFAS began in the late 1940s, and since their genesis, usage has expanded to over 200 categories, from industrial applications to consumer products.² Due to their decades-long, widespread usage, PFAS are detectable in environmental media, wildlife, and humans across the globe.^{3, 4} The most investigated sub-class of PFAS are the perfluoroalkyl acids (PFAAs, often termed “legacy PFAS”).⁵ Of the PFAAs, the two most studied to date are perfluorooctanoic acid (PFOA) and perfluorooctanesulfonic acid (PFOS); however, in recent decades many applications of PFAS have transitioned to using shorter chain replacement PFAS (Glüge et al., 2020; Wang et al., 2017) many of which have been quantified in both abiotic (e.g. drinking water)⁶ and biotic matrices (e.g. plasma and tissue).⁷

The investigation of PFAS in environmental and biological samples relies on both high- and low-resolution mass spectrometry (MS) techniques. While high-resolution MS (HRMS) can be used for identification and quantification of novel compounds, low-resolution MS is the more widespread and affordable approach for studies examining previously identified PFAS and is the primary source of quantitative PFAS measurements. Low-resolution, targeted tandem mass spectrometry (MS/MS) quantification experiments using internal standards are regarded as highly accurate and precise for chemical measurements but still possess the potential for unresolved chemical interference. Investigations of PFAS in biological matrices (e.g., blood, serum, muscle, brain) have identified several instances where compounds interfere with quantitation of PFAS at low-resolution. For example, taurodeoxycholate (a common bile acid) has been observed to mimic the primary perfluorooctanesulfonic acid (PFOS) MS/MS transition (499 → 80),⁸ while endogenous steroid sulfates have been observed to mimic two of the perfluorohexane sulfonic acid (PFHxS) MS/MS transitions (399 → 80 and 399 → 99).⁹ In the case where such interference is anticipated for longer chained PFAS, qualifying ion MS/MS transitions, along with enhanced chemical separation and matched internal standards, can be used to ensure chemical specificity; as a result, the use of qualifying ion MS/MS transitions are suggested to verify the chemical identity for both PFOS and PFHxS. However, short and ultra-short chain PFAS such as perfluorobutanoic acid (PFBA), perfluoropentanoic acid (PFPeA), perfluoropropionic acid (PFPrA) and trifluoroacetic acid (TFA) have limited fragmentation, yielding only one MS/MS transition for quantification and generally no secondary transitions for verification. Positive identification by matched retention time (RT) with an isotopically labeled standard can also be used. However, the relatively high polarity of the short and ultra-short chain PFAS yields early RTs in many chromatographic methods, increasing the risk of misidentifying interfering peaks within the allowed RT window if not carefully scrutinized.

Of note, recent studies undertaken by the FDA and CDC have reported matrix interferences in food and blood for PFBA and PFPeA (Genualdi et al., 2021; Kato et al., 2018), and in 2021, our laboratory identified a saturated oxo-fatty acid (SOFA) in human placenta that interfered in the quantitative ion channel for PFBA (213 → 169) (Bangma et al., 2021). We hypothesized at the time of publication that the SOFA would only be present in biological matrices (e.g., tissue) and not environmental matrices (e.g., surface water). After talking with several collaborators, our laboratory has investigated a number of additional matrices (shellfish, air conditioning (AC) condensate, and hot cocoa mix) that were reported by our collaborators to potentially contain easily measurable levels of PFPeA (263 → 219) or PFBA (213 → 169) when run on low-resolution instrumentation; however, when we investigated those same samples using high resolution instrumentation, we observed matrix interferences in the quantitative ion channel for PFBA and/or PFPeA that could potentially impact the reported levels of these short chain PFAS based on low-resolution measurements. Therefore, the purpose of this study was to identify and report the presence of

PFPeA and PFBA matrix interferences in biological and environmental samples.

2. Materials and methods

2.1. Sample collection

For shellfish analysis, several species were investigated, including Blue mussels (*Mytilus edulis*), Softshell clams (*Mya arenaria*), and Eastern oysters (*Crassostrea virginica*). Shellfish utilized in this investigation were collected in the Fall of 2019 and Fall of 2020 from the Great Bay Estuary in Great Bay, NH. All shellfish were collected at low tide by either rake or hand and stored on ice for transport. Samples were stored at −20 °C in their shells until initial processing. At the time of initial processing, 10 individuals of the same species and similar size were selected to create a pooled or composite sample. Pooled shellfish were homogenized (without shell) using a methanol rinsed metal coffee grinder (Yang et al., 2012; Catherine et al., 2019; Guo et al., 2019). The grinder was cleaned with pure water and methanol between each subsequent pool to prevent any carry over. After initial PFAS analysis, the remaining homogenate was stored at −20 °C until further analysis investigating the potential PFPeA interferent in the Summer of 2021. Altogether, the various composite shellfish samples investigated represented seven different sites from around the Great Bay.

For residential AC condensate PFAS analysis, discharged condensate from nine single-family residential homes were collected in Chapel Hill and Durham, North Carolina during July to August 2021. Samples and DI water field blanks were collected in 500 mL high density polyethylene (HDPE) bottles and temporarily stored in the field with ice packs and a cooler. They were then stored in a refrigerator at 4 °C for five months before processing.

Hot cocoa mix was a commercially available “instant cocoa” packet purchased at a local café.

2.2. Sample preparation

For quantitative validation of PFAS levels, calibration curve solutions were made from purchased PFAC-MXA native stock solution in 100% methanol (Wellington Laboratories, Guelph, ON) which contains both native PFBA and PFPeA. Internal standard (IS) was made using Wellington's labeled MPFBA and M3PFPeA, 50 µg/mL stock solutions. Both native and internal standard solutions were diluted in 100% Optima Liquid Chromatography (LC)/mass spectrometry (MS) grade methanol (Fisher Chemical, Ottawa, ON) to prepare working solutions.

Shellfish and hot cocoa mix samples utilized in this study were spiked with internal standard and extracted using one of three commonly employed solid phase extraction (SPE) methods (Nakayama et al., 2010; Bangma et al., 2018; Kaiser et al., 2021) adapted for the matrices under study. In short, the majority of samples underwent an SPE extraction using Oasis weak anion exchange (WAX) Plus cartridges (225 mg, Waters Corporation) followed by an activated carbon clean-up step using ENVI-Carb SPE cartridges (Supelco); the WAX extraction was either a traditional approach collecting a basic elution fraction, or a modified method with a pooled neutral organic wash and basic elution fraction. In addition to the two WAX extractions, this study also investigated a hydrophilic lipophilic balance (HLB) extraction protocol utilizing Oasis HLB Plus LP Extraction cartridge (Waters Corporation) followed by an activated carbon clean-up step using ENVI-Carb SPE cartridges. Basic outlines for each of the three extraction methods used for shellfish and hot cocoa mix samples can be found in the supplemental information (Tables S1–S2). The extracts were stored at 0 °C prior to analysis, and samples were analyzed within 2 weeks of being processed. Residential AC condensate samples also were extracted using a WAX method (adapted for AC condensate) (Table S3) (Nakayama et al., 2010). In brief, AC condensate samples were concentrated using WAX SPE (no ENV-Carb SPE step required) and isotope dilution. Refer to SI for more

details.

2.3. Liquid chromatography and mass spectrometry analysis

During the initial analysis, suspiciously high PFPeA (up to 83 ng/g PFPeA) was observed in several shellfish samples on a low-resolution instrument. As low molecular weight PFAS typically do not bioaccumulate, authors felt further scrutiny was warranted. To investigate this initial report, we conducted additional low-resolution MS/MS screening for PFPeA in the shellfish using a Thermo Vanquish LC system equipped with a Waters BEH C18 column (2.1 × 50 mm, 1.7 μm) in line with a triple quadrupole (QQQ) Thermo TSQ -Quantis. To match the starting conditions of the initial reporting method, we ran a ramping LC gradient with 2.5 mM Ammonium acetate in 95:5 methanol: de-ionized water for our organic mobile phase and 2.5 mM ammonium acetate in 95:5 de-ionized water: methanol for our aqueous mobile phase, a 50:50 water: methanol starting condition, and 20 μL injection per sample. Throughout the study we utilized alternate ramping LC gradients including a 25:75 gradient starting conditions. Details on all LC solvents and gradients can be found in Table S4. Each 12-min run was followed by 3 min of post-time equilibration at starting conditions (15 min total). Method reporting limits (MRLs) for the QQQ were set based on the lowest point calibration standard with replicate precision <20% RSD and resulted in MRLs of 1 ng/mL for short chain PFAS.

Following the observation of potential short chain PFAS interferences in shellfish, AC condensate, and hot cocoa mix in low-resolution MS/MS, additional analyses were conducted on an HRMS system that employed an Agilent 1200 UHPLC coupled to an Agilent 6546 quadrupole time of flight (QTOF) mass spectrometer with electrospray ionization in negative mode. Details on ESI source parameters for MS1 data collection can be found in Table S5. To match work completed on the low-resolution instrument, chromatographic separation on the Agilent High-resolution system also utilized a Water BEH C18 column (50 mm × 30 mm, 1.7 μm) and various ramping gradients with different starting conditions were investigated. Twenty μL of extract was injected for each sample blank or calibration point. Method reporting limits (MRLs) for the HRMS were set based on the lowest point calibration standard with replicate precision <20% RSD and resulted in MRLs of 4 ng/mL for short chain PFAS.

PFAS in the residential AC condensate samples were quantified by collaborators using a low-resolution AB SCIEX Triple-Quad™ 6500 UHPLC/ESI-MS/MS with multiple reaction monitoring and verified using the HRMS Agilent 6546 quadrupole time of flight mass spectrometer also utilized for the shellfish and cocoa work. The ramping gradient with a 25:75 water: methanol starting condition, as well as the Waters BEH C18 column (50 mm × 30 mm, 1.7 μm), was utilized for both instruments when completing the AC condensate analysis. The PFBA method detection limit (MDL) in AC condensate was calculated using a previously established method (USEPA) resulting in an MDL of 0.28 ng/L.

3. Results and discussion

3.1. PFPeA interferences in shellfish

PFAS investigation in shellfish samples using a ramping LC gradient (starting conditions 50:50 water: methanol) paired with MS/MS on low-resolution instrumentation revealed peaks in the PFPeA ion channel (263 → 219) at or near the retention times of the matching short chain standard (Fig. S1). When the same samples were investigated for PFPeA (263.9757 Da, ± 5 ppm window) using high-resolution instrumentation, no PFPeA was observed in shellfish above the quantitative MRL (4 ng/mL). However, using the same high-resolution instrument and extracting the ion chromatogram with a wide window (symmetric ± 0.5 *m/z*; or in ppm terms ± 500 ppm) captured numerous chromatographic peaks that were close in mass to PFPeA (Fig. 1A–C). Three unique masses were

observed in the shellfish samples within the ± 0.5 *m/z* window at 263.1288 Da, 263.1650 Da, and 263.2016 Da (Fig. S2).

The observed mass of 263.1288 Da was investigated further due to close retention time to the PFPeA standard and the presence of numerous isomeric peaks dispersed across the chromatogram. The precursor mass of the selected unknown was processed using Agilent MassHunter10 to predict an empirical formula for the compound (Elements and limits allowed included C; 3–50, H; 0–120, Cl; 0–10, S; 0–3, F; 0–100, O; 0–10). The highest score predicted formula was assigned as C₁₅H₂₀O₄ for all investigated 263.1288 Da peaks in the chromatogram (Fig. 1D). Prediction scores for each 263.1288 Da peak ranged from 84% to 99% depending on the level of background ions. Lower prediction scores were observed when higher isotope peaks were impacted by the presence of additional shellfish matrix ions, reducing the prediction score certainty (Fig. S3).

Coeluting with the prominent 263.1288 peak in the shellfish chromatogram was a pronounced mass at 131.0607 Da with a distinctive diprotic isotopic signature (0.5 da difference between isotopic peaks) (Fig. 2). In addition, fragmentation of the 236.1288 Da peak yielded a secondary ion at 219.1393 Da that would interfere in the low-resolution PFPeA ion channel (263 → 219) (Fig. 2D). This fragment is the result of a CO₂ elimination similar to the PFPeA monitored 263 → 219 transition. Based on the current evidence relating to the observed chemical formula, the diprotic peak, and fragmentation spectra, we hypothesize that the two largest peaks (if not all of the peaks) in the 263.1288 Da chromatogram represent at least two C₁₅H₂₀O₄ isomers both which have the capability to undergo sequential CO₂ neutral losses.

To investigate probable structures, the assigned formula of C₁₅H₂₀O₄ was searched against the Environmental Protection Agency CompTox Chemicals Dashboard (<https://comptox.epa.gov/dashboard/>) to yield 201 plausible chemical structures (Supplemental File 1), four of which contained two carboxylic acids in their structure. Of the four structures listed in the CompTox dashboard, authors predicted the 263.1288 Da peak has structural similarities to (2-Methylpropyl) (2-phenylethyl) propanedioic acid (CASRN 4361-10-8). Generating predicted spectra of (2-Methylpropyl) (2-phenylethyl)propanedioic acid using CFM-ID (Djoumbou-Feunang et al., 2019) resulted in spectra with many of the same fragments we observed on our instrument except for the prominent 121.1026 Da ([C₉H₁₄]-) fragment (Supplemental File 2). Other potential similar structures not listed on the CompTox dashboard that are likely candidates are unsaturated fatty acids that have undergone ω-oxidation to produce an unsaturated dicarboxylic acid. One possible structure can be observed in Fig. S4. Unfortunately, the number of potential structures for an unsaturated and/or branched dicarboxylic acid with the formula C₁₅H₂₀O₄ is extensive and verifying with a chemical standard is both cost and time prohibitive currently. Therefore, we report a level 3 identification on the Schymanski scale (Schymanski et al., 2014) of an unsaturated dicarboxylic acid. We feel the absolute identification of the interferent is not as important as informing the scientific community of the presence of the analytical interferent.

Further investigation into the prevalence of 263.1288 Da across various species of shellfish using high resolution instrumentation revealed that the unknown masses at 263.1288 Da were predominantly found in oyster and, to a smaller degree, in mussel. Little-to-no signal was observed in clam samples (Fig. S5). To investigate whether various extraction and clean-up methods might impact the interferent at 263.1288 Da, several commonly employed extraction methods were used to prepare a pooled oyster sample and the chromatograms examined for changes in the 263.1288 Da chromatogram (Fig. S6). The results suggest that while none of the methods used in this study will completely eliminate the unknown interfering ions at 263.1288 Da, each extraction method may favor the presence of different isomers.

We also utilized additional LC gradients on low resolution instrumentation to determine if altering the starting conditions of an LC run would improve separation of PFPeA from the identified interfering ions. We began with a 50:50 starting gradient since this was used in our

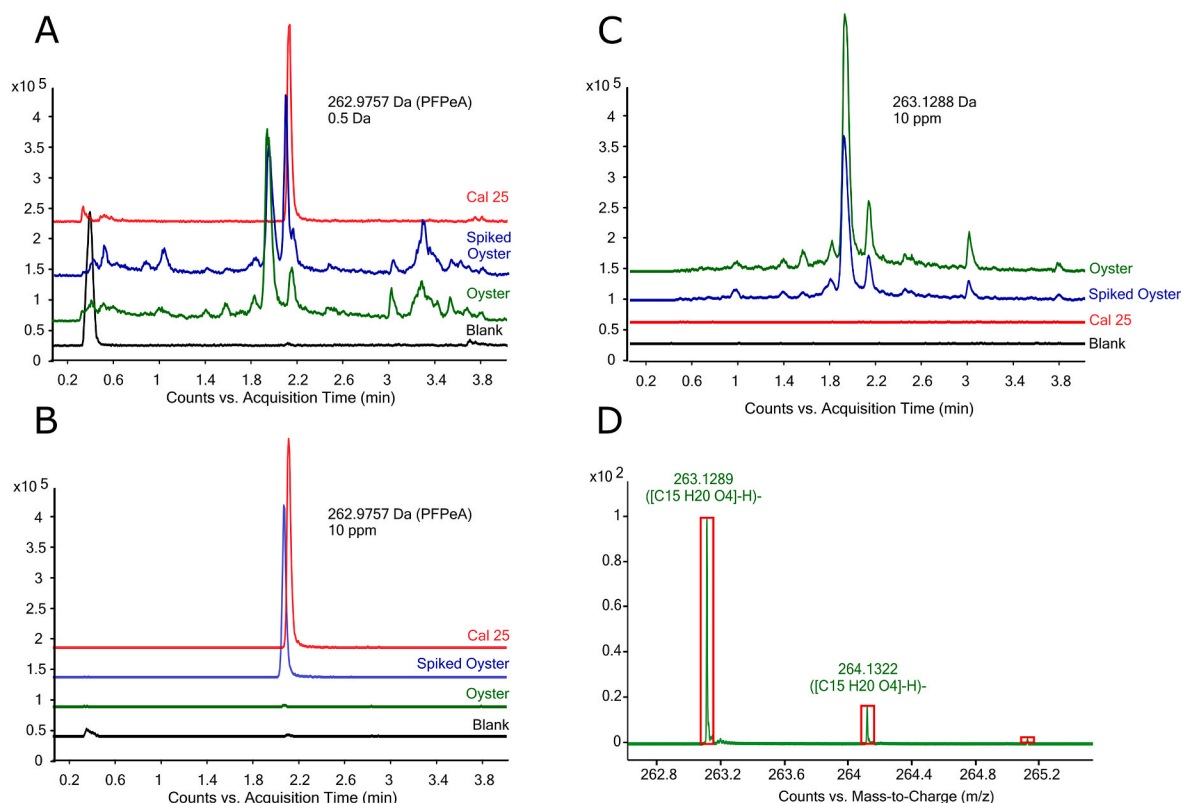


Fig. 1. (A) Wide window extracted ion chromatogram (EIC) for PFPeA (262.9757 Da), (B) narrow window EIC for PFPeA, (C) narrow window EIC for identified unknown ion 263.1288 m/z , and (D) isotopic spectra and formula prediction of 263.1288 m/z . Starting chromatographic conditions were 75:25. (2 columns).

contracting laboratory method and is similar to EPA 537; similar methods are often used to reduce run time and maximize throughput (Fig. S1). Under these conditions we observed the PFPeA interferent eluting close in retention time ($\sim\Delta 0.07$ min) to the PFPeA standard. However, a 75:25 gradient showed improved separation ($\sim\Delta 0.84$ min) making the interfering peak more evident but still possibly misconstrued for PFPeA during automated data processing (Fig. 3). To help verify the adjacent peak, we monitored for the $263 \rightarrow 175$ transition previously observed in the HRMS fragmentation spectra (Fig. 2D). Fragmentation parameters were set to the same parameters used for PFPeA. The resulting peaks suggest that this additional transition can be utilized to monitor for this interfering compound in shellfish samples.

Therefore, we suggest researchers double check PFPeA peaks manually if any study results report either high levels of PFPeA or a high detection frequency of PFPeA. Authors also suggest if high levels or frequency of PFPeA are found in samples, both monitoring for the additional transitions ($263 > 175$ and $263 > 121$) as well as altering the LC conditions may help determine if an interferent is present without the use of HRMS. For samples that are not shellfish-based, authors would highly recommend altering LC method over monitoring for the newly identified transition ($263 \rightarrow 175$) when verifying the presence or absence of PFPeA. Additionally, the exact RT matching of isotope labeled PFPeA and automatically integrated peaks should be carefully scrutinized.

3.2. PFPeA interference in cocoa mix

Recent studies conducted by the FDA have identified a different compound with an isotopic peak that interferes with PFPeA quantitation in food items containing chocolate (Genualdi et al., 2021). Therefore, we investigated a hot cocoa mix with HRMS, and we observed matching masses and fragmentation pattern as had been previously reported ($[M - H]^-$ at 262.1085 Da with a paired ^{13}C isotope $[M - H]^-$ at 263.1120

Da). Formula prediction of the unknown compound resulted in the chemical formula $\text{C}_{14}\text{H}_{18}\text{NO}_4$ with a score of 99.39% (Fig. S7). Similar to the shellfish interference, fragmentation of the hot cocoa interference also revealed a CO_2 loss that would interfere with the PFPeA ion channel. We then verified a 114.0564 Da fragment (Fig. S7) can be monitored as an additional transition ($263 \rightarrow 114$) on low-resolution instrumentation (Fig. S8). This additional transition may help identify the presence or absence of the PFPeA interfering compound that is commonly observed in samples containing chocolate.

3.3. PFBA interference in AC condensate

Our lab continues to monitor for the recently identified SOFA mass that interferes with PFBA quantitation on low-resolution instrumentation (Bangma et al., 2021). One location where SOFA and PFAS both might be expected, is the indoor residential environment. The former is due in part to the presence and oxidation of human skin oils, which together with their oxidation products, partition to indoor air and surfaces (Weschler, 2016; Duncan et al., 2019). The latter is due in part to the presence of PFAS in consumer products (Wang et al., 2017).

Investigation into AC condensate revealed multiple large peaks in the chromatogram for the previously identified SOFA mass at 213.1496 Da (Bangma et al., 2021) suggesting the presence of numerous isomers (Fig. S9). Formula prediction for the five largest peaks in the extracted chromatogram predict the formula $\text{C}_{12}\text{H}_{22}\text{O}_3$ (prediction scores ranged from 96.32% to 99.02%) similar to previous placental work (Bangma et al., 2021). In the placental study, we hypothesized that SOFAs are present only in biological samples and would likely not be present in environmental samples. However, the current results confirm that interferences from SOFAs can be present in environmental samples as well. SOFAs are involved in fatty acid biosynthesis pathways in all eukaryotes, contain a carboxylic acid group for negative mode ionization, and have been documented in bacteria (Ryu et al., 2014), yeast (Kim

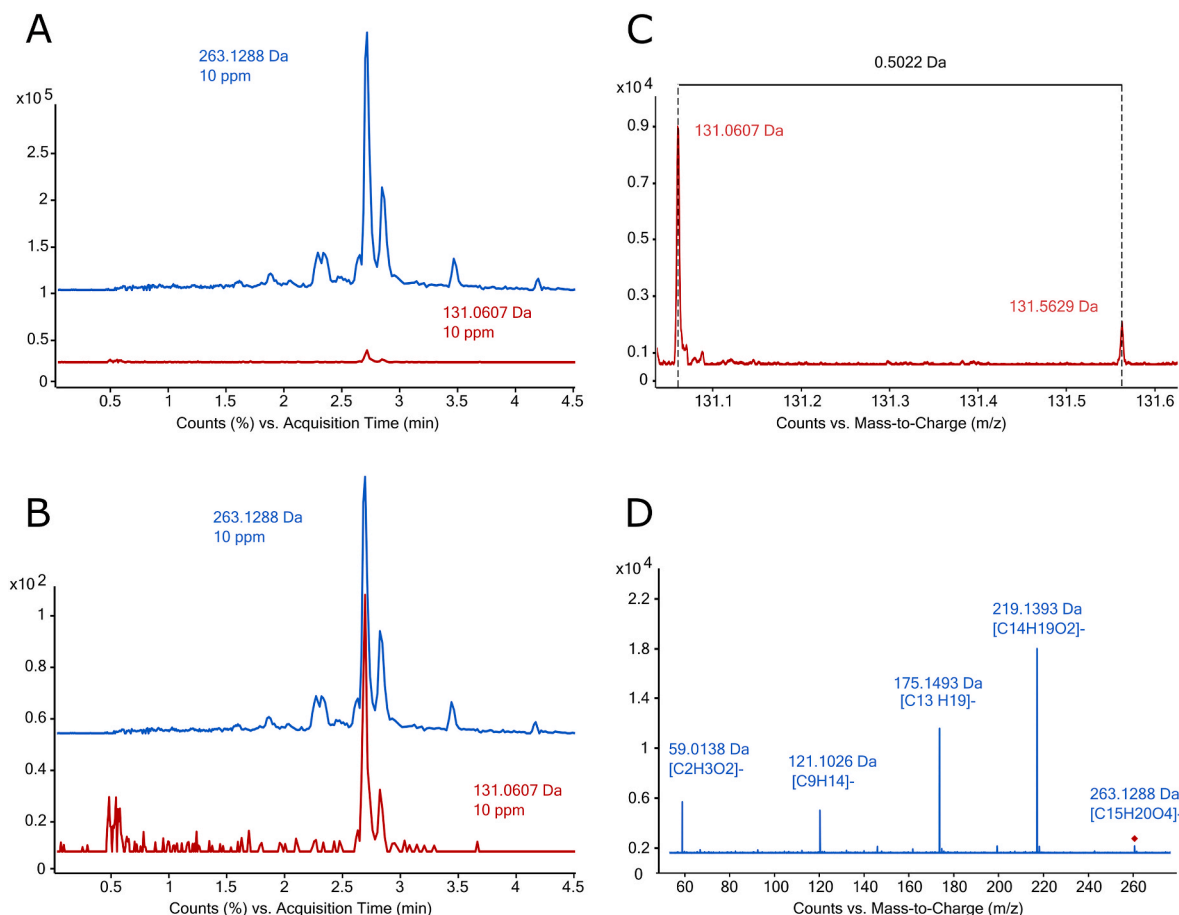


Fig. 2. (A) Shellfish extract narrow window chromatogram of 263.1288 Da and suspected diprotic peak at 131.0607 Da with EIC scaled to the largest peak to maintain scale. (B) Narrow window chromatogram of 263.1288 Da and suspected diprotic peak at 131.0607 Da with both EIC peaks scaled to 100%. (C) Isotopic spectra of diprotic mass. (D) Fragmentation pattern of isolated 263.1288 Da (Average of 5 scans, CID at 10.0, isolation width of 1.3 m/z).

et al., 2007), human breath (Gaugg et al., 2017), human plasma (Batsika, 2021), rabbit plasma (Shestakova et al., 2018), and cows and goats milk (Kokotou et al., 2021). In the indoor environment, the oxidation of skin oils (Wisthaler and Weschler, 2010), cooking oils (Zeng et al., 2020; Köckritz and Martin, 2008), and microorganisms, may all contribute to SOFA signals observed in the AC condensate and can also be expected to be present on indoor surface wipe samples and air samples.

4. Conclusions

Compounds that interfere with quantitation of PFBA and PFPeA and likely other short and ultra-short chain PFAS on low resolution instrumentation are of concern to the scientific community. While both PFPeA and PFBA are likely to be present in the environment, there are no qualifying ions to verify the identity of quantitated peaks on low resolution instrumentation. The lack of qualifying ions may lead to misidentification of interfering peaks from the sample matrix and can result in over reporting of short chain PFAS in the literature. As a part of our investigation, we have identified and confirmed additional transitions for PFPeA interferences in shellfish and hot cocoa mix/chocolate on low-resolution instrumentation. However, we caution that there may be additional naturally-occurring matrix derived interferences for short chain PFAS not observed in this study (and therefore not identified with our additional proposed transitions). As with the non-specific sulfonate transition in PFOS, any compound demonstrating carboxylic acid loss with close enough precursor mass to PFPeA or PFBA has the potential to interfere in the low-resolution ion channel. This is exacerbated by the relatively early elution time and poor separation for short-chain PFAS

when using high-organic gradients commonly used for other legacy PFAAs. Therefore, we suggest investigators utilize chromatographic conditions (similar to the 75:25 conditions utilized in the current study) that permit improved separation from any suspected interfering compounds (Fig. S10) and closely scrutinize the retention time of matched isotope labeled compounds.

Author contributions

Jacqueline Bangma-conceptualization, data curation, methodology, formal analysis, investigation, visualization, writing original draft, James McCord-conceptualization, supervision, visualization, review and editing, Nathan Giffard-resources, visualization, review and editing, Kate Buckman-resources, visualization, review and editing, Jonathan Petali-conceptualization, visualization, review and editing, Celia Chen-resources, visualization, review and editing, Daniel Amparo-resources, data curation, formal analysis, review and editing, Barbara Turpin-resources, review and editing, Glenn Morrison-resources, review and editing, Mark Strynar-conceptualization, supervision, visualization, review and editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

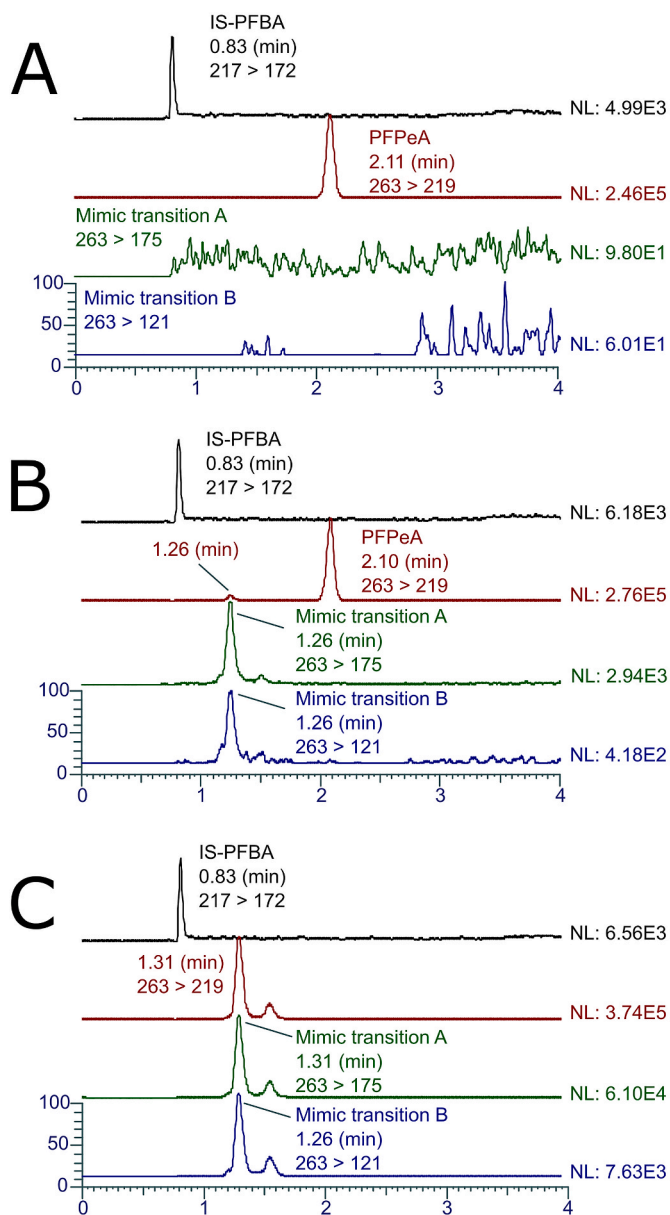


Fig. 3. (A) Calibration standard for PFPeA on a low-resolution instrument, (B) PFPeA spiked oyster sample, and (C) normal oyster sample. Samples shown here were run with a 75:25 L C gradient. Peaks are scaled (%) based on the largest peak in each Extracted Ion Chromatogram (EIC). Height of the tallest peak is listed on the right side of each EIC. NL denotes normalization level for each EIC. IS-PFPeA represents the M3PFPeA isotopically labeled standard from Wellington Laboratory catalogue. (1 or 1.5 columns).

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.chemosphere.2022.137722>.

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